



Free vibration analysis of rotating functionally graded cylindrical shells in thermal environment

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ABSTRACT

The free vibration analysis of rotating functionally graded (FG) cylindrical shells subjected to thermal environment is investigated based on the first order shear deformation theory (FSDT) of shells. The formulation includes the centrifugal and Coriolis forces due to rotation of the shell. The material properties are assumed to be temperature-dependent and graded in the thickness direction. The initial thermo-mechanical stresses are obtained by solving the thermoelastic equilibrium equations. The equations of motion and the related boundary conditions are derived using Hamilton's principle. The differential quadrature method (DQM) as an efficient and accurate numerical tool is adopted to discretize the thermoelastic equilibrium equations and the equations of motion. The convergence behavior of the method is demonstrated and comparison studies with the available solutions in the literature are performed. Finally, the effects of angular velocity, Coriolis acceleration, temperature dependence of material properties, material property graded index and geometrical parameters on the frequency parameters of the FG cylindrical shells with different boundary conditions are investigated.

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1. Introduction

Functionally graded materials (FGMs) are a class of materials in which the material properties are graded but continuous particularly along a given direction. Usually, these materials are made from a mixture of ceramic and metal. FGMs can take the advantage of the desirable properties as heat and corrosion resistance of ceramics and high tensile strength, toughness and bonding capability of metals.

Functionally graded (FG) cylindrical shells are being used as structural components in modern industries of different engineering fields such as aerospace, mechanical and nuclear engineering. Usually, they operate in high temperature environments and in some applications, undergo rotations with constant angular velocities. The induced thermo-mechanical initial stresses by the thermal environment and centrifugal forces together with the temperature dependence of the material properties can change their vibration characteristics to be become different from those of stationary FG shells in non-thermal environment. Hence, for the purpose of their engineering design and manufacture, the accurate evaluation of their vibration characteristics under these working conditions becomes essential.

The free vibration analyses of rotating isotropic and laminated composite cylindrical shells have been studied extensively in previous works (see for example Refs. [1–7]). Also, the elastic or thermoelastic analyses of rotating FG cylindrical shells have been investigated by some researchers [8–15]. On the other hand, the free vibration behaviors of FG cylindrical shells with and without thermal environment effects have been investigated in the recent years [16–25]. However, to the best of authors' knowledge, there is no research works on the free vibration analysis of the rotating FG cylindrical shells with and without thermal environment effects.

Due to the practical importance of the free vibration of the rotating FG cylindrical shells under thermal environment and also lack of information in the open literature in this regards, this problem is studied in the present work. For this purpose, firstly, the initial thermo-mechanical stresses due to the thermal environment and rotation of the shell are obtained by solving the thermoelastic equilibrium equations, which is derived based on the first order shear deformation theory (FSDT) of shells. Then, the equations of motion and the related boundary conditions for the free vibration of the shell around its thermoelastic equilibrium position are derived using Hamilton's principle in conjunction with the FSDT of shells. The material properties are assumed to be temperature-dependent and graded in the radial direction, which can vary according to power law distributions in terms of the volume fractions of the constituents. The differential quadrature method (DQM) as an efficient and accurate numerical tool [3,15,21,26–30] is employed to discretize the thermoelastic equilibrium equations

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and the free vibration equations of motion subjected to the related boundary conditions. After demonstrating the correctness of the formulation together with the convergence and accuracy of the method of solution, the effects of angular velocity, Coriolis acceleration, temperature dependence of material properties, material property graded index, thickness and length-to-mean radius ratios on the frequency parameters are studied.

2. Governing equations

Consider an FG cylindrical shell with mean radius R , thickness h and length L , which rotates about its axis with a constant angular velocity Ω (Fig. 1). Cylindrical coordinate system with coordinate variables (x, θ, z) is used to label the material points of the FG shell in the undeformed reference configuration.

2.1. Temperature-dependent FGMs relations

The material properties of the shell vary continuously and smoothly in the thickness direction of the shell. Without loss of generality of the formulation and the method of solution, in this study the effective material properties are obtained using the power law distribution,

$$P(z, T) = P_M(T) + [P_C(T) - P_M(T)] \left(\frac{2z + h}{2z} \right)^p \quad (1)$$

where P denotes a generic material property, $P_C(T)$ and $P_M(T)$ are the corresponding values at the outer surface (ceramic rich) and inner surface (metal rich) of the shell; p is the power law index (or the material property graded index), which is a positive real number; T is the temperature (in Kelvin) at an arbitrary material point of the cylinder. The temperature dependence of a typical material property 'Q' can be represented as a function of temperature as [31],

$$Q(T) = Q_0(Q_{-1}T^{-1} + 1 + Q_1T + Q_2T^2 + Q_3T^3) \quad (2)$$

The coefficients Q_i ($i = -1, 0, 1, 2, 3$) are unique to the constituent materials.

2.2. Evaluation of the initial equilibrium configuration and thermo-mechanical stresses

The difference between the thermal conditions of the operational and manufacturing environments can adjust the initial temperature

distribution in the FG shell, which together with the mechanical constraints at the boundaries and non-homogeneity of the material properties can induce some thermal stresses in the FG shell. These initial thermal stresses in conjunction with the initial mechanical stresses due to rotation of the FG shell, affect the vibration behavior of the FG shells. In addition, the free vibration takes occur around this deformed configuration. Hence, before performing the vibration analysis of the FG shells under these operational conditions, the initial thermoelastic equilibrium state and the related thermo-mechanical stresses should be determined. For this purpose, the temperature distribution within the shell should be evaluated firstly. It is assumed that the shell is stress free at the temperature T_0 and the temperature rise is uniform or varies across its thickness and no heat generation source exists within it. Without loss of generality, the thermal boundary conditions of the FG shell are assumed to the prescribed temperature at its inner and outer surfaces. Hence, the temperature distribution along the thickness direction can be obtained by solving the following steady state one-dimensional heat transfer equation in the cylinder,

$$\frac{d}{dz} \left[K(z, T) \frac{dT}{dz} \right] = 0; \quad T = T_M \text{ at } z = -h/2 \text{ and } T = T_C \text{ at } z = h/2 \quad (3)$$

where K is the thermal conductivity of the shell. Using the polynomial series solution, one obtains,

$$T(z) = T_M + \frac{(T_C - T_M)}{\hat{C}} \left[\eta - \frac{K_{CM}}{(p+1)K_M} \eta^{p+1} + \frac{K_{CM}^2}{(2p+1)K_M^2} \eta^{2p+1} - \frac{K_{CM}^3}{(3p+1)K_M^3} \eta^{3p+1} + \frac{K_{CM}^4}{(4p+1)K_M^4} \eta^{4p+1} - \frac{K_{CM}^5}{(5p+1)K_M^5} \eta^{5p+1} \right] \quad (4)$$

where

$$\hat{C} = 1 - \frac{K_{CM}}{(p+1)K_M} + \frac{K_{CM}^2}{(2p+1)K_M^2} - \frac{K_{CM}^3}{(3p+1)K_M^3} + \frac{K_{CM}^4}{(4p+1)K_M^4} - \frac{K_{CM}^5}{(5p+1)K_M^5}, \quad K_{CM} = K_C - K_M, \quad \eta = \frac{2z + h}{2z} \quad (5a-c)$$

Due to axisymmetric temperature distribution and also the centrifugal forces, the nonzero displacement components at an arbitrary material point of the shell $(\bar{u}_0, \bar{w}_0)$ in the thermoelastic equilibrium state can be approximated in the thickness direction according to the FSDT of shells as,

$$\bar{u}_0(x, z) = u_0(x) + z\varphi_0^x(x), \quad \bar{w}_0(x, z) = w_0(x) \quad (6a, b)$$

where u_0 and w_0 are the displacements of an arbitrary material point on the mid-surface of the shell along the x and z -directions,

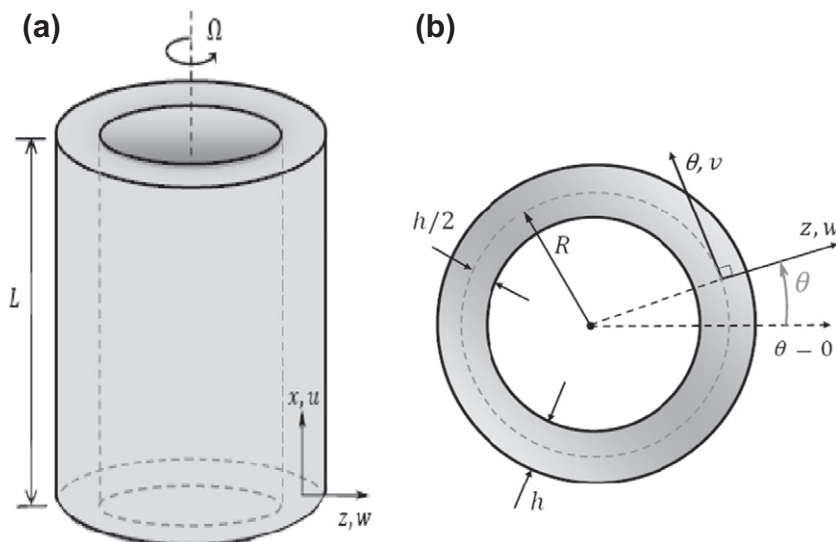


Fig. 1. The geometry of the rotating FG cylindrical shell.

respectively; also φ_0^x is the bending rotation of the cross section of the shell about the θ -axis. One can realize that the tangential component of the displacement is zero in this state. Hereafter, a subscript '0' is used to represent the variables of the deformation field and the stress components in the equilibrium state.

The thermoelastic equilibrium equations and the related boundary conditions can be derived in terms of the displacement and the rotation components by using the principle of virtual work as,

Equilibrium equations:

$$\delta u_0: A_1^n \frac{d^2 u_0}{dx^2} + \bar{A}_0^n \frac{dw_0}{dx} + B_1^n \frac{d^2 \varphi_0^x}{dx^2} = \frac{dN_T}{dx} \quad (7)$$

$$\delta w_0: -\bar{A}_0^n \frac{du_0}{dx} + A_1^m \frac{d^2 w_0}{dx^2} - A_{-1}^n w_0 + (A_1^m - \bar{B}_0^n) \frac{d\varphi_0^x}{dx} = -I_{11} \Omega^2 (R + w_0) - P_T \quad (8)$$

$$\delta \varphi_0^x: B_1^n \frac{d^2 u_0}{dx^2} + (B_0^n - A_1^m) \frac{dw_0}{dx} + D_1^n \frac{d^2 \varphi_0^x}{dx^2} - A_1^m \varphi_0^x = \frac{dM_T}{dx} \quad (9)$$

Boundary conditions at the ends $x = 0$ and $x = L$:

$$\text{Either } \delta u_0 = 0 \text{ or } A_1^n \frac{du_0}{dx} + \bar{A}_0^n w_0 + B_1^n \frac{d\varphi_0^x}{dx} = N_T \quad (10a, b)$$

$$\text{Either } \delta w_0 = 0 \text{ or } A_1^m \left(\frac{dw_0}{dx} + \varphi_0^x \right) = 0 \quad (11a, b)$$

$$\text{Either } \delta \varphi_0^x = 0 \text{ or } B_1^n \frac{du_0}{dx} + \bar{B}_0^n w_0 + D_1^n \frac{d\varphi_0^x}{dx} = M_T \quad (12a, b)$$

In above equations, the stiffness coefficients, the resultant thermal stresses and the inertia term are, respectively

$$\begin{bmatrix} A_i^n & B_i^n & D_i^n \\ \bar{A}_i^n & \bar{B}_i^n & \bar{D}_i^n \end{bmatrix} = \int_{-\frac{h}{2}}^{\frac{h}{2}} \bar{E}(z, T) \begin{bmatrix} 1 \\ z \\ z^2 \end{bmatrix} \begin{bmatrix} 1 \\ z \\ z^2 \end{bmatrix} (R + z)^i dz, \quad (13a, b)$$

$$\begin{bmatrix} A_i^m & B_i^m & D_i^m \\ \bar{A}_i^m & \bar{B}_i^m & \bar{D}_i^m \end{bmatrix} = \int_{-\frac{h}{2}}^{\frac{h}{2}} G(z, T) \begin{bmatrix} k_s \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ z \\ z^2 \end{bmatrix} (R + z)^i dz \text{ for } i = -1, 0, 1$$

$$[N_T \ M_T] = \int_{-\frac{h}{2}}^{\frac{h}{2}} (1 + \nu) \bar{E}(z, T) \varepsilon_T(z) [1 \ z] (R + z) dz,$$

$$P_T = \int_{-\frac{h}{2}}^{\frac{h}{2}} (1 + \nu) \bar{E}(z, T) \varepsilon_T dz,$$

$$\varepsilon_T = \int_{T_0}^T \alpha(T, z) dT, \quad I_{11} = \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho(R + z) dz \quad (14a-d)$$

where $\bar{E}(z, T) = \frac{E(z, T)}{1 - \nu^2(z, T)}$; $E(z, T)$ is the Young's modulus; $\nu(z, T)$ is the Poisson's ratio; $G(z, T) = \frac{E}{2(1 + \nu)}$ is the shear rigidity; k_s is the shear

Table 1
Temperature-dependent coefficients of material properties for ceramic (ZrO₂) and metals (Ti–6Al–4V) [31].

Material	Q_{-1}	Q_0	Q_1	Q_2	Q_3
E Ti–6Al–4V	0	122.7 (GPa)	-4.605×10^{-4}	0	0
ZrO ₂	0	132.2 (GPa)	-3.805×10^{-4}	-6.127×10^{-8}	0
ν Ti–6Al–4V	0	0.2888	1.108×10^{-4}	0	0
ZrO ₂	0	0.3330	0	0	0
ρ Ti–6Al–4V	0	4420 (kg/m ³)	0	0	0
ZrO ₂	0	3657 (kg/m ³)	0	0	0
α Ti–6Al–4V	0	7.43×10^{-6} (1/K)	7.483×10^{-4}	-3.621×10^{-7}	0
ZrO ₂	0	13.3×10^{-6} (1/K)	-1.421×10^{-3}	9.549×10^{-7}	0
K Ti–6Al–4V	0	6.10 (W/mK)	0	0	0
ZrO ₂	0	1.78 (W/mK)	0	0	0

Table 2

Convergence of the frequency parameters of the clamped rotating FG cylindrical shell [$h/R = 0.05, L/R = 2.5, p = 1, \Delta T = 800$ (K)].

m	N_x	$\Omega = 500$ (rad/s)			$\Omega = 1000$ (rad/s)		
		λ_1	λ_2	λ_3	λ_1	λ_2	λ_3
1	7	0.4392	0.7371	0.8711	0.4153	0.7510	0.9108
	9	0.4391	0.7369	0.8768	0.4147	0.7439	0.9310
	11	0.4392	0.7372	0.8765	0.4146	0.7451	0.9262
	13	0.4392	0.7372	0.8765	0.4147	0.7452	0.9266
	15	0.4392	0.7372	0.8765	0.4147	0.7452	0.9266
2	7	0.3091	0.5663	0.7548	0.3585	0.6398	0.8334
	9	0.3088	0.5651	0.7739	0.3578	0.6244	0.8665
	11	0.3090	0.5658	0.7738	0.3576	0.6262	0.8613
	13	0.3090	0.5658	0.7738	0.3577	0.6263	0.8615
	15	0.3090	0.5658	0.7738	0.3577	0.6263	0.8615

Table 3

Convergence of the frequency parameters of the soft simply supported rotating FG cylindrical shell [$h/R = 0.05, L/R = 2.5, p = 1, \Delta T = 800$ (K)].

m	N_x	$\Omega = 500$ (rad/s)			$\Omega = 1000$ (rad/s)		
		λ_1	λ_2	λ_3	λ_1	λ_2	λ_3
1	9	0.4015	0.5447	0.7034	0.3610	0.5662	0.6812
	11	0.4015	0.5447	0.7042	0.3615	0.5665	0.6839
	13	0.4016	0.5448	0.7035	0.3617	0.5666	0.6832
	17	0.4017	0.5448	0.7036	0.3618	0.5667	0.6833
	19	0.4017	0.5448	0.7036	0.3618	0.5667	0.6833
2	9	0.2351	0.5139	0.6945	0.2740	0.5490	0.7289
	11	0.2349	0.5154	0.6889	0.2746	0.5430	0.7249
	13	0.2350	0.5143	0.6947	0.2749	0.5420	0.7281
	17	0.2353	0.5144	0.6948	0.2751	0.5422	0.7284
	19	0.2353	0.5144	0.6948	0.2751	0.5422	0.7284

Table 4

Accuracy of the frequency parameters of an infinitely long rotating cylindrical shell ($h/R = 0.002, L/R = 40, \nu = 0.3$).

m	$\Omega = 0.5$ (rad/s)		$\Omega = 2.5$ (rad/s)		$\Omega = 5$ (rad/s)	
	Present	Analytic [1]	Present	Analytic [1]	Present	Analytic [1]
2	0.00163	0.00162	0.00200	0.00196	0.00245	0.00246
3	0.00443	0.00444	0.00464	0.00467	0.00496	0.00500
4	0.00842	0.00844	0.00856	0.00862	0.00898	0.00887
5	0.01359	0.01362	0.01375	0.01377	0.0142	0.0140
6	0.01994	0.01996	0.02010	0.02008	0.0205	0.0202
7	0.02744	0.02746	0.02760	0.02756	0.0281	0.0277

Table 5

Accuracy of the natural frequencies (Hz) for a soft simply supported FG cylindrical shell ($h/R = 0.05, L/R = 20$).

p	m	Present	Loy [16]
0	1	12.898	12.917
	2	31.543	31.603
	3	87.977	88.267
1	1	13.195	13.234
	2	32.269	32.418
	3	90.007	90.569
2	1	13.305	13.344
	2	32.514	32.683
	3	90.689	91.309

correction factor; ε_T is the thermal strain; $\alpha(z, T)$ is the thermal expansion coefficient; and $\rho(z, T)$ is the mass density of the FG shell.

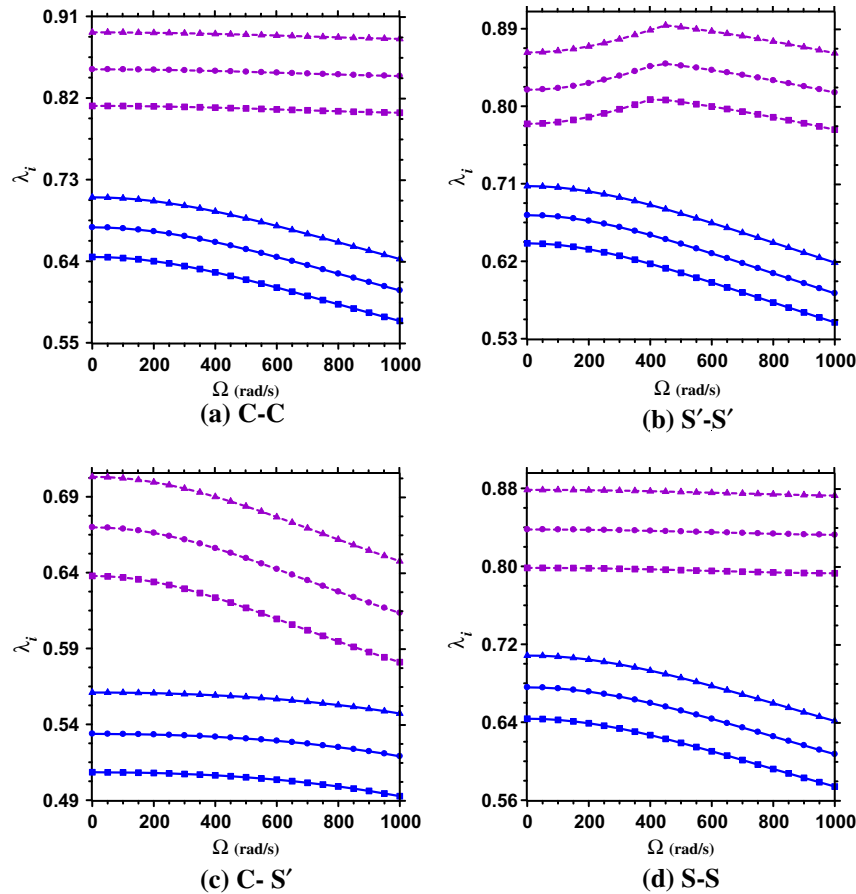


Fig. 2. The influence of the power law index together with the angular velocity on the axisymmetric frequency parameters of the FG rotating cylindrical shells ($h/R = 0.05, L/R = 2.5, m = 0, N_x = 19$): (—) $i = 1$, (---) $i = 2$, (Δ) $p = 0$, ($\bullet$) $p = 1$, ($\blacksquare$) $p = 4$.

Different types of the classical boundary conditions on the ends $x = 0$ and $x = L$ can be obtained by combining the conditions stated in Eqs. (10)–(12). For example,

Hard simply support (S):

$$u_0 = 0, w_0 = 0, B_1^n \frac{d^2 u_0}{dx^2} + (B_0^n - A_1^m) \frac{dw_0}{dx} + D_1^n \frac{d^2 \varphi_0^x}{dx^2} - A_1^m \varphi_0^x = \frac{dM_T}{dx} \quad (15a-c)$$

Soft simply support (S'):

$$w_0 = 0, A_1^n \frac{d^2 u_0}{dx^2} + \bar{A}_0^n \frac{dw_0}{dx} + B_1^n \frac{d^2 \varphi_0^x}{dx^2} = \frac{dN_T}{dx}, \quad B_1^n \frac{d^2 u_0}{dx^2} + (B_0^n - A_1^m) \frac{dw_0}{dx} + D_1^n \frac{d^2 \varphi_0^x}{dx^2} - A_1^m \varphi_0^x = \frac{dM_T}{dx} \quad (16a-c)$$

Clamped (C):

$$u_0 = 0, w_0 = 0, \varphi_0^x = 0 \quad (17a-c)$$

2.3. Free vibration analysis

It is assumed that the free vibration of the shell occurs about its thermo-mechanical equilibrium configuration. The displacement components due to this motion are measured from this position. Hence, based on the FSDT shell theory, the total displacement components ($\hat{u}, \hat{v}, \hat{w}$) at an arbitrary material point (x, θ, z) along the x, θ

and z -directions measured from the shell un-deformed configuration become, respectively,

$$\begin{aligned} \hat{u}(x, \theta, z, t) &= \bar{u}_0(x, z) + u(x, \theta, t) + z\varphi^x(x, \theta, t), \\ \hat{v}(x, \theta, z, t) &= v(x, \theta, t) + z\varphi^\theta(x, \theta, t), \\ \hat{w}(x, \theta, z, t) &= w_0(x) + w(x, \theta, t) \end{aligned} \quad (18a-c)$$

where $\varphi^x(x, \theta, t)$ and $\varphi^\theta(x, \theta, t)$ are the bending rotation of the cross section of the shell about θ - and x -axis, respectively.

The equations of motion together with the related boundary conditions can be obtained using Hamilton's principle, which is,

$$\int_{t_1}^{t_2} (\delta K - \delta U) dt = 0 \quad (19)$$

In the above equation, t_1 and t_2 are two arbitrary times; δ is the variational operator; K and U are the kinetic and potential energies of the FG shell. The variation of the potential energy of the FG shell, which includes the virtual work of the initial thermo-mechanical stresses, can be written as

$$\begin{aligned} \delta U &= \int_0^{2\pi} \int_0^L \int_{-\frac{h}{2}}^{\frac{h}{2}} [(\sigma_{xx} + \sigma_{0xx})\delta\hat{\epsilon}_{xx} + (\sigma_{\theta\theta} + \sigma_{0\theta\theta})\delta\hat{\epsilon}_{\theta\theta} \\ &\quad + (\sigma_{xz} + \sigma_{0xz})\delta\hat{\gamma}_{xz} + \sigma_{x\theta}\delta\hat{\gamma}_{x\theta} + \sigma_{\theta z}\delta\hat{\gamma}_{\theta z}](R + z)dz dx d\theta \end{aligned} \quad (20)$$

where $\sigma_{ij}(i, j = x, \theta, z)$ are the dynamic parts of the total stress tensor components due to the shell vibratory motion; σ_{0xx} , $\sigma_{0\theta\theta}$ and σ_{0xz} are the only nonzero initial thermo-mechanical stress components; $\hat{\epsilon}_{ii}$ and $\hat{\gamma}_{ij}$ are the normal and shear components of Green's strain

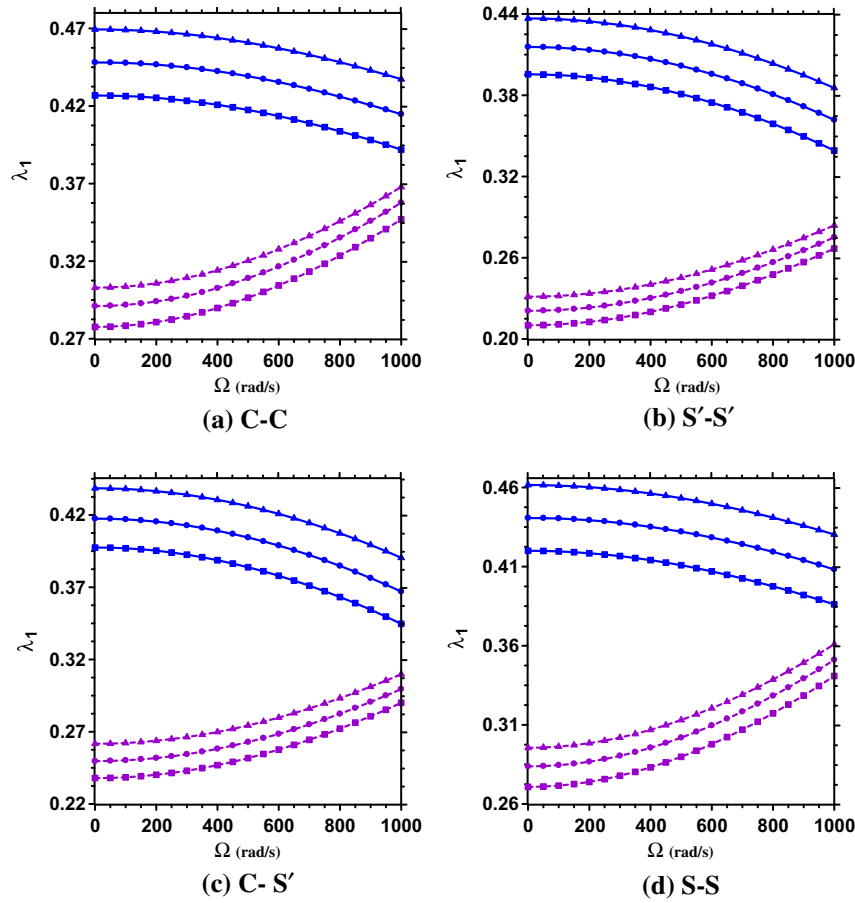


Fig. 3. The influence of the power law index together with the angular velocity on the asymmetric fundamental frequency parameters of the FG rotating cylindrical shells ($h/R = 0.05, L/R = 2.5, N_x = 19$): (—) $m = 1$, (---) $m = 2$, (Δ) $p = 0$, ($\bullet$) $p = 1$, ($\blacksquare$) $p = 4$.

tensor, which are presented in the [Appendix A](#). To include the effects of the initial thermo-mechanical stresses in the equations of motion, the nonlinear terms in the strain–displacement relations should be considered, however this means not that nonlinear free vibration is considered [28].

The variation of the kinetic energy of the FG shell can be written as,

$$\delta K = \int_0^{2\pi} \int_0^L \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho \left[\frac{\partial \hat{u}}{\partial t} \frac{\partial \delta \hat{u}}{\partial t} + \frac{\partial \hat{v}}{\partial t} \frac{\partial \delta \hat{v}}{\partial t} + \frac{\partial \hat{w}}{\partial t} \frac{\partial \delta \hat{w}}{\partial t} + \Omega \left(\hat{w} \frac{\partial \delta \hat{v}}{\partial t} + \frac{\partial \hat{v}}{\partial t} \delta \hat{w} - \frac{\partial \hat{w}}{\partial t} \delta \hat{v} - \hat{v} \frac{\partial \delta \hat{w}}{\partial t} \right) + \Omega^2 (\hat{v} \delta \hat{v} + \hat{w} \delta \hat{w}) \right] \times (R + z) dz dx d\theta \quad (21)$$

Inserting Eqs. (20) and (21) into Eq. (19) and using the constitutive relations together with the thermoelastic equilibrium Eqs. (7)–(9), one obtains the equations of motion and the related boundary conditions as follows,

Equations of motion:

$$\delta u : (A_1^n + N_{0xx}) \frac{\partial^2 u}{\partial x^2} + (\bar{A}_{-1}^m + T_{000}) \frac{\partial^2 u}{\partial \theta^2} + (\bar{A}_0^m + \bar{A}_0^n) \frac{\partial^2 v}{\partial \theta \partial x} + \bar{A}_0^n \frac{\partial w}{\partial x} + (B_1^n + M_{0xx}) \frac{\partial^2 \varphi^x}{\partial x^2} + 2N_{0xz} \frac{\partial \varphi^x}{\partial x} + (S_{000} + \bar{B}_{-1}^m) \frac{\partial^2 \varphi^x}{\partial \theta^2} + \frac{dN_{0xz}}{dx} \varphi^x + (\bar{B}_0^n + \bar{B}_0^m) \frac{\partial^2 \varphi^0}{\partial \theta \partial x} = I_{11} \frac{\partial^2 u}{\partial t^2} + I_{12} \frac{\partial^2 \varphi^x}{\partial t^2} \quad (22)$$

$$\delta v : (\bar{A}_0^n + \bar{A}_0^m) \frac{\partial^2 u}{\partial \theta \partial x} + (\bar{A}_1^m + N_{0xx}) \frac{\partial^2 v}{\partial x^2} + (A_{-1}^n + T_{000}) \frac{\partial^2 v}{\partial \theta^2} - (A_{-1}^m + T_{000}) v + (A_{-1}^m + A_{-1}^n + 2T_{000}) \frac{\partial w}{\partial \theta} + (\bar{B}_0^n + \bar{B}_0^m) \frac{\partial^2 \varphi^x}{\partial \theta \partial x} + (\bar{B}_1^m + M_{0xx}) \frac{\partial^2 \varphi^0}{\partial x^2} + 2N_{0xz} \frac{\partial \varphi^0}{\partial x} + (B_{-1}^n + S_{000}) \frac{\partial^2 \varphi^0}{\partial \theta^2} + (A_0^m - B_{-1}^m - S_{000} \frac{dN_{0xz}}{dx}) \varphi^0 = I_{11} \frac{\partial^2 v}{\partial t^2} + I_{12} \frac{\partial^2 \varphi^0}{\partial t^2} + 2I_{11} \Omega \frac{\partial w}{\partial t} - I_{11} \Omega^2 v - I_{12} \Omega^2 \varphi^0 \quad (23)$$

$$\delta w : -\bar{A}_0^n \frac{\partial u}{\partial x} - (A_{-1}^n + A_{-1}^m + 2T_{000}) \frac{\partial v}{\partial \theta} + (A_1^m + N_{0xx}) \frac{\partial^2 w}{\partial x^2} + (A_{-1}^m + T_{000}) \frac{\partial^2 w}{\partial \theta^2} - (A_{-1}^n + T_{000}) w + (A_1^m - \bar{B}_0^n) \frac{\partial \varphi^x}{\partial x} + (A_0^m - B_{-1}^m - B_{-1}^n - 2S_{000}) \frac{\partial \varphi^0}{\partial \theta} = I_{11} \frac{\partial^2 w}{\partial t^2} - 2I_{11} \Omega \frac{\partial v}{\partial t} - 2I_{12} \Omega \frac{\partial \varphi^0}{\partial t} - I_{11} \Omega^2 w \quad (24)$$

$$\delta \varphi^x : S_{000} \frac{\partial^2 u}{\partial \theta^2} + (\bar{B}_0^n + \bar{B}_0^m) \frac{\partial^2 u}{\partial x \partial \theta} + (\bar{B}_1^m + M_{0xx}) \frac{\partial^2 v}{\partial x^2} + B_{-1}^n \frac{\partial^2 v}{\partial \theta^2} + (A_0^m - B_{-1}^m) v + (B_{-1}^n + B_{-1}^m - A_0^m) \frac{\partial w}{\partial \theta} + (\bar{D}_0^n + \bar{D}_0^m) \frac{\partial^2 \varphi^x}{\partial x \partial \theta} + E_{000} \frac{\partial^2 \varphi^x}{\partial \theta^2} + (\bar{D}_1^m + F_{0xx}) \frac{\partial^2 \varphi^0}{\partial x^2} + \frac{dF_{0xx}}{dx} \frac{\partial \varphi^0}{\partial x} + D_{-1}^n \frac{\partial^2 \varphi^0}{\partial \theta^2} + (2B_0^m - D_{-1}^m - A_0^m + \frac{dM_{0xz}}{dx}) \varphi^0 = I_{12} \frac{\partial^2 u}{\partial t^2} + I_{13} \frac{\partial^2 \varphi^x}{\partial t^2} \quad (25)$$

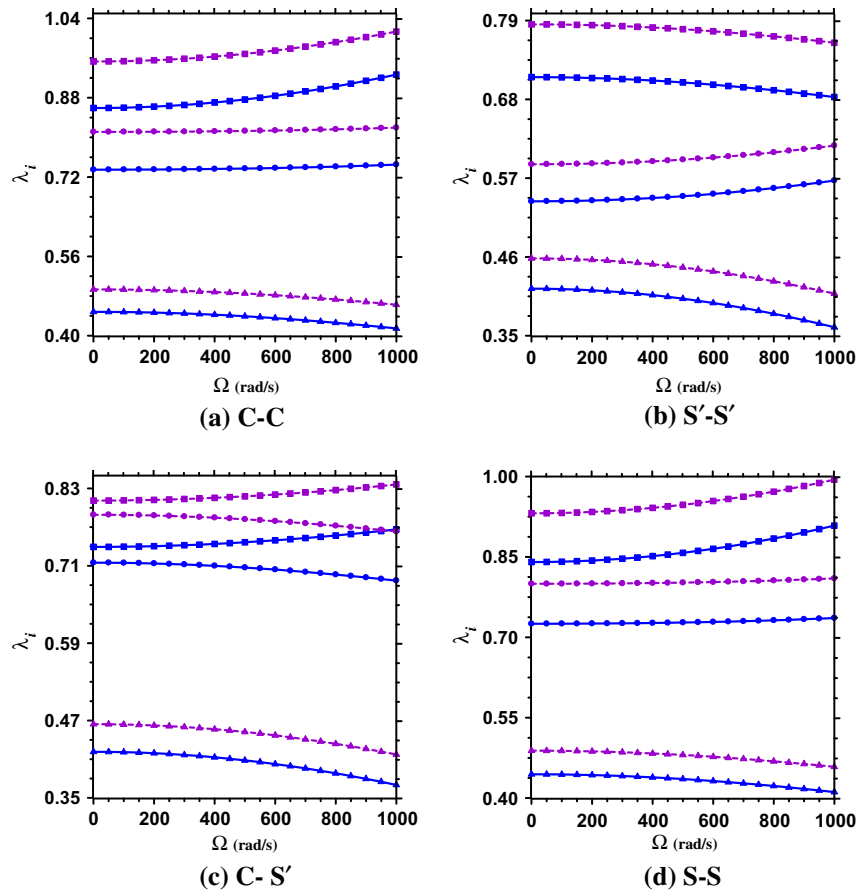


Fig. 4. The influence of the temperature dependence of material properties on the frequency parameters of the rotating FG cylindrical shell ($h/R = 0.05, L/R = 2.5, p = 1, N_x = 19, m = 1$): (—) Temperature-dependent, (---) temperature-independent, (▲) $i = 1$, (●) $i = 2$, (■) $i = 3$.

$$\begin{aligned}
 \delta \varphi^\theta : & (B_1^n + M_{0xx}) \frac{\partial^2 u}{\partial x^2} + \bar{B}_{-1}^m \frac{\partial^2 u}{\partial \theta^2} + (\bar{B}_0^n + \bar{B}_0^m) \frac{\partial^2 v}{\partial x \partial \theta} + S_{0\theta\theta} \frac{\partial^2 v}{\partial \theta^2} \\
 & - S_{0\theta\theta} v + (B_0^n - A_1^m) \frac{\partial w}{\partial x} + 2S_{0\theta\theta} \frac{\partial w}{\partial \theta} + (D_1^n + F_{0xx}) \frac{\partial^2 \varphi^x}{\partial x^2} \\
 & + \left(\frac{dF_{0xx}}{dx} \right) \frac{\partial \varphi^x}{\partial x} + \bar{D}_{-1}^m \frac{\partial^2 \varphi^x}{\partial \theta^2} - \left(A_1^m - \frac{dM_{0xz}}{dx} \right) \varphi^x \\
 & + (\bar{D}_0^n + \bar{D}_0^m) \frac{\partial^2 \varphi^\theta}{\partial x \partial \theta} + E_{0\theta\theta} \frac{\partial^2 \varphi^\theta}{\partial \theta^2} - E_{0\theta\theta} \varphi^\theta \\
 & = I_{12} \frac{\partial^2 v}{\partial t^2} + I_{13} \frac{\partial^2 \varphi^\theta}{\partial t^2} + 2I_{12}\Omega \frac{\partial w}{\partial t} - I_{12}\Omega^2 v - I_{13}\Omega^2 \varphi^\theta \quad (26)
 \end{aligned}$$

Boundary conditions at the ends $x = 0$ and $x = L$:

Either $\delta u = 0$ or

$$\begin{aligned}
 (A_1^n + N_{0xx}) \frac{\partial u}{\partial x} + \bar{A}_0^n \frac{\partial v}{\partial \theta} + \bar{A}_0^n w + (B_1^n + M_{0xx}) \frac{\partial \varphi^x}{\partial x} \\
 + N_{0xz} \varphi^x + \bar{B}_0^n \frac{\partial \varphi^\theta}{\partial \theta} = 0 \quad (27a, b)
 \end{aligned}$$

Either $\delta v = 0$ or

$$\begin{aligned}
 \bar{A}_0^m \frac{\partial u}{\partial \theta} + (\bar{A}_1^m + N_{0xx}) \frac{\partial v}{\partial x} + \bar{B}_0^m \frac{\partial \varphi^x}{\partial \theta} + (\bar{B}_1^m + M_{0xx}) \frac{\partial \varphi^\theta}{\partial x} \\
 + N_{0xz} \varphi^\theta = 0 \quad (28a, b)
 \end{aligned}$$

Either $\delta w = 0$ or

$$(A_1^m + N_{0xx}) \frac{\partial w}{\partial x} + A_1^m \varphi^x = 0 \quad (29a, b)$$

Either $\delta \varphi^x = 0$ or

$$\begin{aligned}
 (B_1^n + M_{0xx}) \frac{\partial u}{\partial x} + \bar{B}_0^n \frac{\partial v}{\partial \theta} + \bar{B}_0^n w + (D_1^n + F_{0xx}) \frac{\partial \varphi^x}{\partial x} + M_{0xz} \varphi^x + \bar{D}_0^n \frac{\partial \varphi^\theta}{\partial \theta} \\
 = 0 \quad (30a, b)
 \end{aligned}$$

Either $\delta \varphi^\theta = 0$ or

$$\begin{aligned}
 (\bar{B}_1^m + M_{0xx}) \frac{\partial v}{\partial x} + \bar{B}_0^m \frac{\partial v}{\partial \theta} + \bar{D}_0^m \frac{\partial \varphi^x}{\partial \theta} + (\bar{D}_1^m + F_{0xx}) \frac{\partial \varphi^\theta}{\partial x} + M_{0xz} \varphi^\theta \\
 = 0 \quad (31a, b)
 \end{aligned}$$

where

$$\begin{aligned}
 \left[\begin{Bmatrix} N_{0xx} \\ N_{0xz} \end{Bmatrix} \begin{Bmatrix} M_{0xx} \\ M_{0xz} \end{Bmatrix} \begin{Bmatrix} F_{0xx} \\ F_{0xz} \end{Bmatrix} \right] &= \int_{-\frac{h}{2}}^{\frac{h}{2}} \begin{Bmatrix} \sigma_{0xx} \\ \sigma_{0xz} \end{Bmatrix} [1 \quad z \quad z^2] (R+z) dz \\
 [T_{0\theta\theta} \quad S_{0\theta\theta} \quad E_{0\theta\theta}] &= \int_{-\frac{h}{2}}^{\frac{h}{2}} \sigma_{0\theta\theta} [1 \quad z \quad z^2] \frac{dz}{(R+z)}, \quad (I_{12}, I_{13}) \\
 &= \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho(z, z^2) (R+z) dz \quad (32a-c)
 \end{aligned}$$

Similar to the case of the thermoelastic equilibrium analysis, different types of the classical boundary conditions on the ends $x = 0$ and $x = L$ can be obtained by combining the conditions stated in Eqs. (27)–(31) for the free vibration analysis.

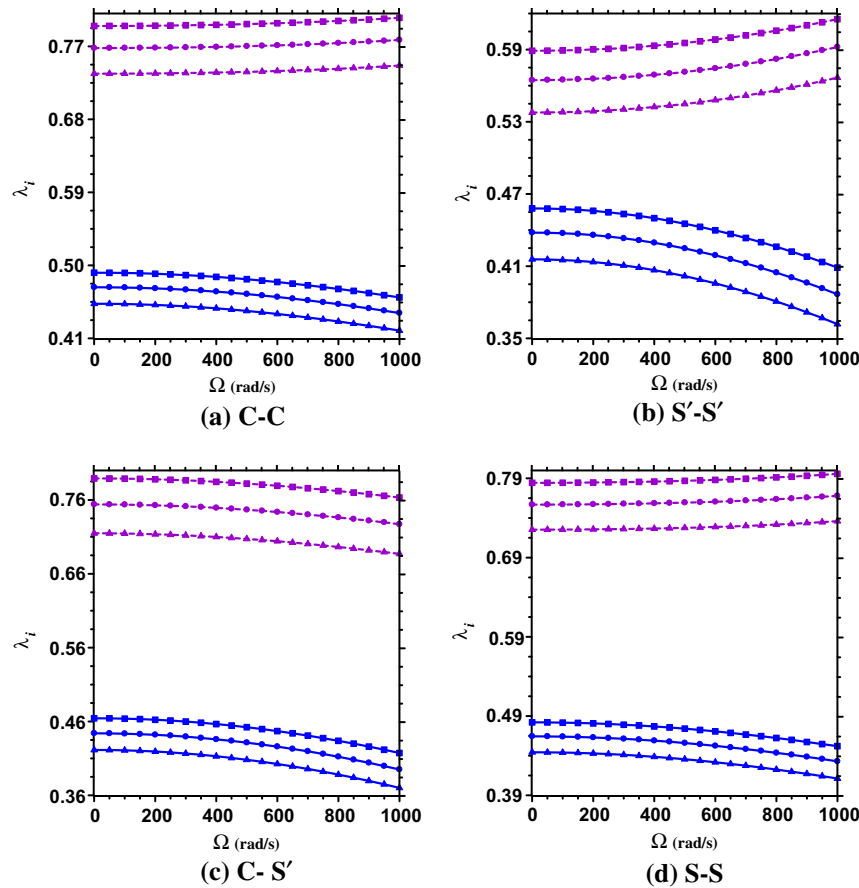


Fig. 5. The influences of temperature rise on the frequency parameters of the rotating FG cylindrical shells ($L/R = 2.5, h/R = 0.05, p = 1, m = 1, N_x = 19$): (—) $i = 1$, (---) $i = 2$, (Δ) $\Delta T = 0$ (K), ($\bullet$) $\Delta T = 400$ (K), ($\blacksquare$) $\Delta T = 800$ (K).

Considering the geometrical periodicity of the cylindrical shell, the displacement and rotation components for the free vibration analysis can be expressed as

$$\begin{aligned} \begin{Bmatrix} u(x, \theta, t) \\ w(x, \theta, t) \\ \varphi^x(x, \theta, t) \end{Bmatrix} &= \begin{Bmatrix} U_m(x, t) \\ W_m(x, t) \\ \Phi_m^x(x, t) \end{Bmatrix} \cos(m\theta), \quad \begin{Bmatrix} v(x, \theta, t) \\ \varphi^\theta(x, \theta, t) \end{Bmatrix} \\ &= \begin{Bmatrix} V_m(x, t) \\ \Phi_m^\theta(x, t) \end{Bmatrix} \sin(m\theta) \end{aligned} \quad (33a, b)$$

where $m(=0, 1, 2, \dots, \infty)$ is the circumferential wave number of the shell. Inserting these expressions into the governing equations and the related boundary conditions, one obtains a set one-dimensional differential equation of motion.

3. Solution procedure

Due to coupling of the governing equations, if it is not impossible to obtain an exact solution for them, it is difficult to find such a solution. Hence, an approximate method should be used to get the numerical results.

The differential quadrature method (DQM) as an efficient and accurate numerical tool [3,15,21,26–30] is employed to solve both the thermoelastic equilibrium equations as well as the equations of motion. The basic idea of the DQM is that the derivative of a function with respect to a coordinate variable at a given grid point is approximated as the linear weighted sums of its values at all of the grid points in the domain of that variable. At the domain and boundary grid points, the strong forms of the equations of motion and the boundary conditions are discretized using the DQ-

cretization rules, respectively. Here, for brevity purpose, only the DQ discretized form of Eq. (7) is presented. Based on this method, at a grid point i with $i = 2, \dots, N_x - 1$, one gets

$$A_1^n \sum_{k=1}^{N_x} B_{ik}^x u_{0k} + \bar{A}_0^n \sum_{k=1}^{N_x} A_{ik}^x w_{0k} + B_1^n \sum_{k=1}^{N_x} B_{ik}^x \varphi_{0k}^x = \left(\frac{dN_T}{dx} \right)_i \quad (34)$$

where A_{ik}^x and B_{ik}^x represent the weighting coefficients of the first and second order derivatives along the x -direction, respectively [21,26–30]. In a similar manner, the discretized form the other thermoelastic equilibrium equations and the related boundary conditions transformed into a system of linear algebraic equations, which can be solved using the conventional system of algebraic equations solver. But for the free vibration analysis, one obtains a system of linear ordinary differential equations. In order to reduce the computational efforts for solving this system of differential equations, the boundary and domain degrees of freedom are separated. In vector forms, they are denoted as $\{b\}$ and $\{d\}$, respectively. Based on these definitions, the DQ discretized form of the equations of motion and the related boundary conditions can be represented in the matrix form as, respectively,

$$[M]\{\ddot{d}\} + [C]\{\dot{d}\} + [K_{db}]\{b\} + [K_{dd}]\{d\} = \{0\} \quad (35)$$

$$[K_{bb}]\{b\} + [K_{bd}]\{d\} = \{0\} \quad (36)$$

where $(\dot{*}) = \frac{d}{dt} (*)$ and $(\ddot{*}) = \frac{d^2}{dt^2} (*)$. Also, the elements of the stiffness matrices $[K_{di}]$ ($i = b, d$), the damper matrix $[C]$ and the mass matrix $[M]$ are obtained from the discretized form of the equations of motion and those of the stiffness matrixes $[K_{bi}]$ ($i = b, d$) are obtained from the discretized form of boundary conditions.

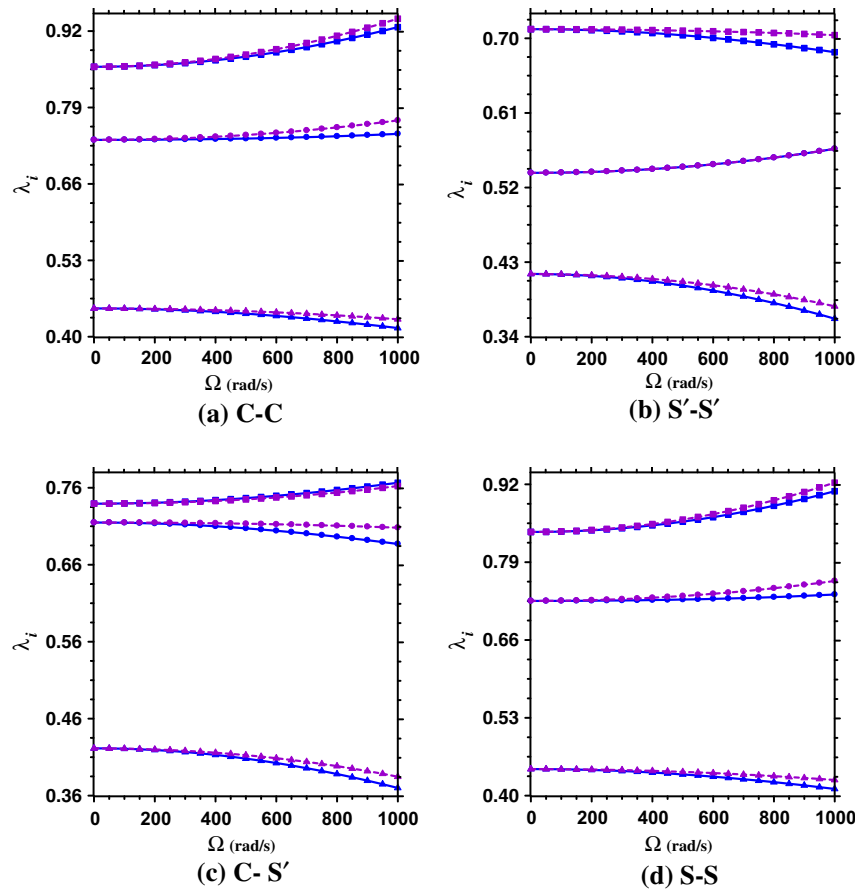


Fig. 6. The effect of the Coriolis acceleration on the frequency parameters of the rotating FG cylindrical shells ($L/R = 2.5, h/R = 0.05, p = 1, m = 1, N_x = 19$): (—) With Coriolis acceleration, (---) Without Coriolis acceleration, (Δ) $i = 1$, ($\bullet$) $i = 2$, ($\blacksquare$) $i = 3$.

Eliminating the boundary degrees of freedom from Eq. (35) using Eq. (36), the result reads

$$[M]\{\ddot{d}\} + [C]\{\dot{d}\} + [\bar{K}]\{d\} = \{0\} \quad (37)$$

where $[\bar{K}] = [K_{dd}] - [K_{db}][K_{bb}]^{-1}[K_{bd}]$.

Using the state space method, the equations of motion can be rearranged as

$$[A]\{\dot{q}\} + [B]\{q\} = \{0\} \quad (38)$$

where $\{q\} = \begin{Bmatrix} \{d\} \\ \{\dot{d}\} \end{Bmatrix}$ is the new degrees of freedom vector and

$$[A] = \begin{bmatrix} [0] & [M] \\ [I] & [0] \end{bmatrix}, \quad [B] = \begin{bmatrix} [\bar{K}] & [C] \\ [0] & -[I] \end{bmatrix} \quad (39a, b)$$

Using the relation $\{q(t)\} = \{Q\}e^{st}$, in which s is a complex number and $\{Q\}$ is the vector of degrees of freedom amplitude, Eq. (38) is transformed into a standard eigenvalue problem as

$$(s[A] + [B])\{Q\} = \{0\} \quad (40)$$

Solving the system of eigenvalue Eq. (40), the natural frequencies are obtained. It should be mentioned that the imaginary parts of the eigenvalues of Eq. (40) are the natural frequency (ω_m) of the rotating FG cylindrical shell.

4. Numerical results

In this section, firstly, the fast rate of convergence of the method for evaluating the frequency parameters of the FG rotating cylin-

drical shells in thermal environment is investigated. Also, its accuracy is demonstrated by performing the comparison studies with the available results in the literature. Otherwise specified, the FG material properties are assumed to vary according to the power law distribution (1) and the non-dimensional natural frequency parameters are defined as

$$\lambda_i = \omega_{mi} R \sqrt{\frac{\rho_M (1 - \nu_{OM}^2)}{E_{OM}}} \quad (41)$$

In the solved examples, it is assumed that $T_M = T_0 = 300$ (K) and $T_C = T_0 + \Delta T$ (K); also, if not stated, a value of $\Delta T = 800$ (K) is used. The material properties of Ti-6Al-4V and ZrO_2 , as given in Table 1, are used to obtain the new numerical results, which are chosen from the work of Kim [31]. They are valid for the temperature range of $300 \text{ K} \leq T \leq 1100 \text{ K}$. Also, the shear correction factor is assumed to be $k_s = 5/(6 - \nu)$, which is chosen from Ref. [32].

As a first example, the convergence of the non-dimensional natural frequency parameters of the clamped-clamped (C-C) and soft simply-soft simply supported (S'-S') FG rotating cylindrical shells subjected to thermal environment are presented in Tables 2 and 3, respectively. The results are prepared for two different values of the angular velocity and the circumferential wave number (m). In all cases, the fast rate of convergence of the method is quite evident. It is also clear that a number of nineteen grid points along the x -direction ($N_x = 19$) are sufficient to obtain the results with at least four significant digits convergence.

In order to validate the formulation and the method of solution, in Table 4 the results for the free vibration of isotropic rotating

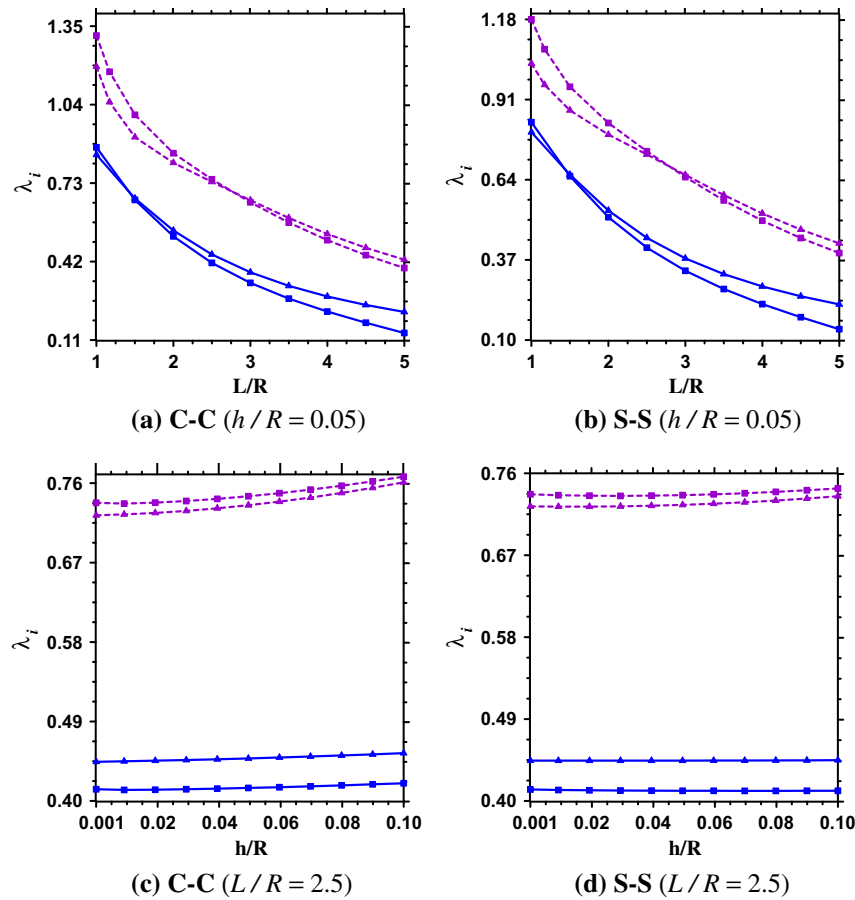


Fig. 7. The influence of the length and thickness-to-mean radius ratios on the frequency parameters of the rotating FG cylindrical shells ($p = 1, m = 1, N_x = 19$): (—) $i = 1$, (---) $i = 2$, (▲) $\Omega = 0$ (rad/s), (■) $\Omega = 1000$ (rad/s).

cylindrical shells with soft simply supported ends are compared with the analytical solution of Chen et al. [1]. They presented an analytical solution for the free vibration of a very long isotropic rotating cylindrical shell with soft simply supported ends. For the purpose of comparison, non-dimensional natural frequency parameters are obtained from Eq. (45) of their work [1], which is

$$\omega_m = \frac{2m\Omega}{m^2 + 1} + \sqrt{\frac{m^2(m^2 - 1)^2}{m^2 + 1} \frac{E_{0M}h^2}{12R^2\rho_M(1 - \nu^2)} + \frac{m^4 + 3}{(m^2 + 1)^2} (1 - \nu^2)\Omega^2} \quad (42)$$

A value of ($L/R = 40$) is used for the length-to-mean radius ratio to model a long shell using the present formulation. The results are prepared for the three different values of the angular velocity and in all cases, excellent agreement is observed between the results of the two approaches.

As another comparison study to verify the accuracy of the approach for the analysis of FG cylindrical shells, the natural frequencies (Hz) of a soft simply supported non-rotating FG cylindrical shell are compared with those obtained by Loy et al. [16] in Table 5. They employed Love's shell theory in conjunction with the Rayleigh–Ritz method to obtain the natural frequencies. To find such a solution, they assumed that all the material properties vary according to the power law distribution based on Eq. (1). The material properties of the constituents are as follows [16]

$$E_M = 207.788 \text{ GPa}, \quad \nu_M = 0.318, \quad \rho_M = 8166 \text{ (kg/m}^3\text{)}; \\ E_C = 205.098 \text{ GPa}, \quad \nu_C = 0.31, \quad \rho_C = 8900 \text{ (kg/m}^3\text{)}$$

The results are presented for the three different values of the circumferential wave number (m) and the material property graded index (p). Again, one can see that excellent agreement exists between the results of the two approaches. In addition, it is obvious that the results of the present approach are slightly smaller than those of Loy et al. [16], which more point out the correctness of the solutions. This is because the present approach is based on the FSDT shell theory and includes the transverse shear deformations, which consequently reduce the overall stiffness of the shell with respect to the classical shell theory that used by Loy et al. [16].

After demonstrating the convergence and accuracy of the approach, some parametric studies for the frequency parameters of the rotating FG cylindrical shells with four different types of the end boundary conditions under the thermal environment are presented.

As a first example, the influences of the shell angular velocity together with the material property graded index (p) on the frequency parameters of the axisymmetric ($m = 0$) and asymmetric ($m = 1, 2$) modes of the rotating FG cylindrical shells are shown in Figs. 2 and 3, respectively. It is interesting to note that for all types of boundary conditions under consideration, by increasing the angular velocity, the first frequency parameters associated to the circumferential wave numbers ($m = 0, 1$) decreases but those of the circumferential wave number ($m = 2$) increases. However, for the axisymmetric case ($m = 0$), its effect on the second frequency parameter depends on the end boundary conditions of the shells. In all cases, the frequency parameters decrease when increasing the material property graded index (p).

The effect of temperature dependence of material properties on the frequency parameters of the rotating FG cylindrical shells are

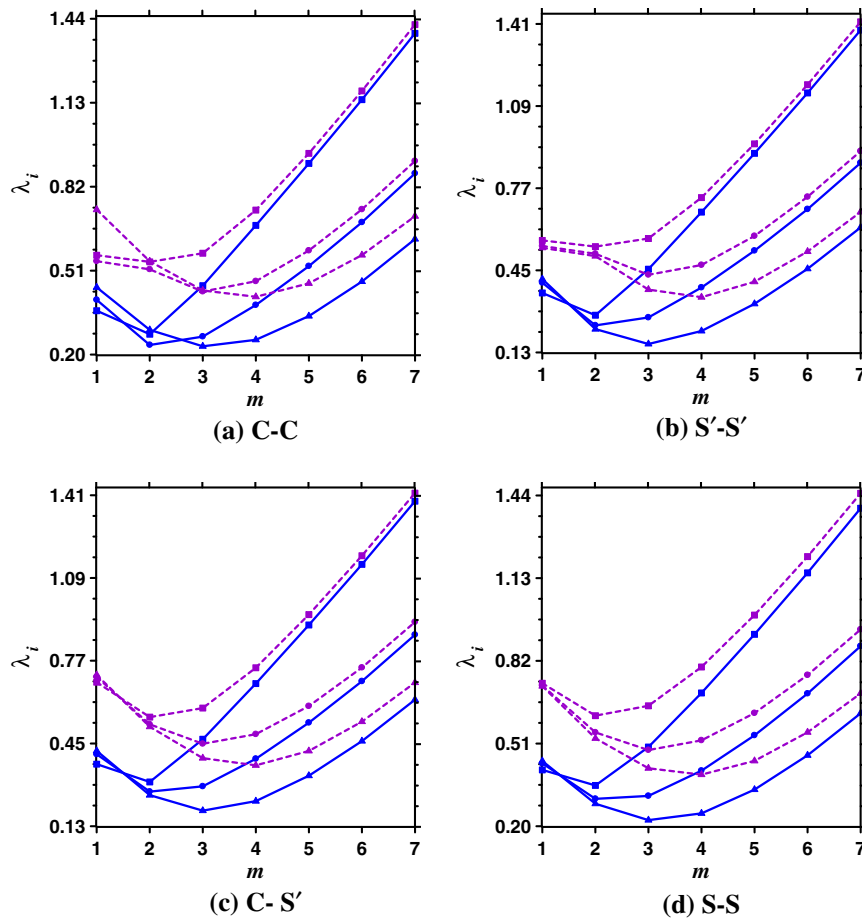


Fig. 8. Comparison of the frequency parameters of the rotating FG cylindrical shells for different circumferential wave numbers ($h/R = 0.05, L/R = 2.5, p = 1, N_x = 19$): (—) $i = 1$, (---) $i = 2$, (Δ) $\Omega = 0$ (rad/s), ($\bullet$) $\Omega = 500$ (rad/s), ($\blacksquare$) $\Omega = 1000$ (rad/s).

studied in Fig. 4. It is observable that the temperature dependence of material properties has significant effect on the frequency parameters and considerably reduces them. Also, the influence of temperature rise, as an important thermal parameter, on the frequency parameters is presented in Fig. 5. It is obvious that the temperature rise has a contrary effect on the frequency parameters, i.e., the frequency parameters reduce by increasing this parameter. This is due to the fact that the shell stiffness decreases by increasing the temperature rise and consequently the frequency parameters of the shell decrease.

The effect of Coriolis acceleration on the frequency parameters of the rotating FG shells subjected to different boundary conditions is investigated in Fig. 6. It can be seen that increasing the angular velocity, the impact of Coriolis acceleration on the frequency parameters increases.

The variations of the frequency parameters against the length and thickness-to-mean radius ratios of the rotating FG shells, as the two important geometrical parameters, for the rotating and non-rotating shells are presented in Fig. 7. From Fig. 7a and b, it can be seen that the length-to-mean radius ratio has significant effect on the frequency parameters. Also, it is observable from Fig. 7c and d that the thickness-to-mean radius ratio has little effect on the first frequency parameters and its effect on the second frequency parameters depend on the shell boundary conditions.

A comparison between the first two frequency parameters associated with the different circumferential wave number (m) of the rotating FG cylindrical shells for the three different values of the angular velocity are exhibited in Fig. 8. It is found that the first two frequency parameters of a given circumferential wave number

approaches to each other when the angular velocity becomes large. Also, it is interesting to note that in the most cases under consideration, the minimum values of the frequency parameters occurs at the circumferential wave number other than ($m = 1$).

5. Conclusion

Based on the first order shear deformation theory of shells, the free vibration analysis of the rotating FG cylinder with temperature-dependent and radially graded material properties in thermal environment was investigated. The thermoelastic equilibrium equations for evaluating the initial thermo-mechanical stresses due to the thermal environment and centrifugal forces are obtained. Then, by considering the Coriolis acceleration and the initial thermo-mechanical stresses, the free vibration equations of motion around the thermoelastic equilibrium state were derived. The governing equations were solved using the differential quadrature method. After demonstrating the convergence and accuracy of the presented formulation and the method of solution, the parametric studies were carried out and based on the presented numerical results, the following conclusions can be made.

The increase of angular velocity can either increase or decrease the frequency parameters of a rotating FG shell with prescribed geometrical and material parameters. This depends on the ends boundary conditions and also the circumferential wave numbers of the shell. Also, it was found that the first two frequency parameters related to a given circumferential wave number, approaches to each other when the angular velocity becomes large. Furthermore, it could be observed that the temperature dependence of the material

properties, temperature rise, and increase of the material properties graded index reduce the frequency parameters of the FG shells. Regards to the influence of Coriolis acceleration on the frequency parameters, it was shown that it has little effect at low angular velocity and its effect increases when the angular velocity increases. The length-to-mean radius ratio has significant effect on the frequency parameters but the thickness-to-mean radius ratio has little effect on them. Finally, it was shown that the minimum value of the asymmetric frequency parameter can occur for a circumferential wave number other than ($m = 1$), which depend on the boundary conditions at ends of the FG shells and its the angular velocity.

Appendix A

Based on the three-dimensional elasticity theory, the nonlinear strain displacement relations in the cylindrical coordinate system become [33]

$$\begin{aligned}\hat{\epsilon}_{xx} &= \frac{\partial \hat{u}}{\partial x} + \frac{1}{2} \left[\left(\frac{\partial \hat{u}}{\partial x} \right)^2 + \left(\frac{\partial \hat{v}}{\partial x} \right)^2 + \left(\frac{\partial \hat{w}}{\partial x} \right)^2 \right], \\ \hat{\epsilon}_{\theta\theta} &= \frac{1}{(R+z)} \left(\frac{\partial \hat{v}}{\partial \theta} + \hat{w} \right) + \frac{1}{2(R+z)^2} \left[\left(\frac{\partial \hat{w}}{\partial \theta} - \hat{v} \right)^2 + \left(\frac{\partial \hat{v}}{\partial \theta} + \hat{w} \right)^2 + \left(\frac{\partial \hat{u}}{\partial \theta} \right)^2 \right], \\ \hat{\epsilon}_{zz} &= \frac{\partial \hat{w}}{\partial z} + \frac{1}{2} \left[\left(\frac{\partial \hat{u}}{\partial z} \right)^2 + \left(\frac{\partial \hat{v}}{\partial z} \right)^2 + \left(\frac{\partial \hat{w}}{\partial z} \right)^2 \right], \\ \hat{\gamma}_{z\theta} &= \frac{\partial \hat{v}}{\partial z} + \frac{1}{(R+z)} \left(\frac{\partial \hat{w}}{\partial \theta} - \hat{v} \right) + \frac{1}{(R+z)} \left[\frac{\partial \hat{w}}{\partial z} \left(\frac{\partial \hat{w}}{\partial \theta} - \hat{v} \right) + \frac{\partial \hat{v}}{\partial z} \left(\frac{\partial \hat{v}}{\partial \theta} + \hat{w} \right) + \frac{\partial \hat{u}}{\partial z} \frac{\partial \hat{u}}{\partial \theta} \right], \\ \hat{\gamma}_{\theta x} &= \frac{\partial \hat{v}}{\partial x} + \frac{\partial \hat{u}}{(R+z)\partial \theta} + \frac{1}{(R+z)} \left[\frac{\partial \hat{w}}{\partial x} \left(\frac{\partial \hat{w}}{\partial \theta} - \hat{v} \right) + \frac{\partial \hat{v}}{\partial z} \left(\frac{\partial \hat{v}}{\partial \theta} + \hat{w} \right) + \frac{\partial \hat{u}}{\partial \theta} \frac{\partial \hat{u}}{\partial x} \right], \\ \hat{\gamma}_{zx} &= \frac{\partial \hat{w}}{\partial x} + \frac{\partial \hat{u}}{\partial z} + \frac{\partial \hat{u}}{\partial x} \frac{\partial \hat{u}}{\partial z} + \frac{\partial \hat{v}}{\partial x} \frac{\partial \hat{v}}{\partial z} + \frac{\partial \hat{w}}{\partial x} \frac{\partial \hat{w}}{\partial z}\end{aligned}\quad (\text{A.1-6})$$

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